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工业环境电缆的选择与截面确定
Cable selection and sizing in an industrial environment

工业电缆的选型包括导体材料、芯数、绝缘、铠装及护套；导体截面积则须同时满足安装后的载流量、压降、故障保护和损耗要求。相同负载在不同安装方式及散热条件下，可能需要不同的电缆规格。

Cable selection covers conductor material, number of cores, insulation, armour and sheath. Conductor size must satisfy installed current-carrying capacity, voltage drop, fault protection and loss requirements together. The same load may require different cables under different installation and heat-dissipation conditions.

1 三组表的关系 / How the three table groups work together

表组 / Table group	作用	Function
Appendix 7 安装方式	先识别现场的敷设、支承、间距及围护条件，确定相应 Method 编号。	Identify the actual installation, support, spacing and enclosure conditions and the corresponding Method number.
Table A6(1) 载流量及压降	按适用电缆类型、截面、相制及 Method 栏位，读取基准载流量和压降数据。	Read the base capacity and voltage-drop data for the cable type, cross-section, phase arrangement and Method column.
Tables A5(1)–A5(6) 修正系数	按温度、成组、隔热、保护器件及地沟条件，采用适用的修正及表注。	Apply the relevant temperature, grouping, insulation, protective-device and trench corrections, including their notes.

2 负载及电缆基本选型 / Load and basic cable selection

因素 / Factor	定性解释	Qualitative explanation
Design current (Ib) 设计电流	按工业设备实际负载、最大需量、同时使用情况及工作周期确定，并考虑功率因数和效率。电动机启动还须核查短时电流、启动压降及保护特性。	Establish Ib from actual plant loads, maximum demand, simultaneous operation and duty cycle, allowing for power factor and efficiency. Motor circuits also need checks of starting current, starting voltage drop and protective-device characteristics.
Cable construction 电缆结构	铜与铝、单芯与多芯、PVC 与 XLPE，以及铠装和护套均影响适用表格及环境性能。电缆额定电压也须适合供电系统；接线端子的温度和连接要求同样需要满足。	Conductor material, core construction, insulation, armour and sheath govern the applicable rating data and environmental suitability. The voltage rating must suit the system, and terminal temperature and connection requirements must also be met.
Loaded conductors 载流导体及排列	须区分单相、三相和直流，以及表内 2 根或 3／4 根导体条件。平面排列、三叶排列、接触或分隔敷设会改变散热及交流电抗。非线性负载的中性导体电流亦须评估。	Distinguish single-phase, three-phase and d.c. circuits and the stated two- or three/four-conductor arrangements. Flat, trefoil, touching and spaced arrangements affect cooling and a.c. reactance. Neutral current from nonlinear loads also requires assessment.

3 环境温度及成组修正 / Ambient temperature and grouping

因素 / Factor	定性解释	Qualitative explanation
Table A5(1) 一般温度修正	环境越热，导体可用的温升余量越小，通常须降低载流量或增大截面。按绝缘种类、运行温度及实际最高环境温度选择 Ca。仅作短路保护时查 A5(1)；兼作过载保护时，除 BS 3036 半封闭熔断器外也采用 A5(1)。	Higher ambient temperature reduces the available temperature-rise margin, normally requiring derating or a larger conductor. Select Ca for the insulation, operating temperature and expected ambient temperature. A5(1) applies to short-circuit-only protection and also to overload protection, except where a BS 3036 semi-enclosed fuse is used.
Table A5(2) 半封闭熔断器的温度修正	当 BS 3036 半封闭熔断器承担过载保护时，应按 A5(2)选择温度修正。温度表的选择与 A5(5)的保护器件类型修正应分别考虑，并满足各自适用条件。	When a BS 3036 semi-enclosed fuse provides overload protection, select its temperature correction from A5(2). Selection of the temperature table and the protective-device correction in A5(5) are separate decisions, each subject to its stated conditions.
Table A5(3) 成组数量及敷设排列	多回路同时带载会互相加热。Cg 取决于回路或多芯电缆数量、围护、单层排列、桥架方向及接触或分隔情况。例如表中围护或成束情况，2 回路为 0.80，4 回路为 0.65，说明相同截面可用载流量随成组增加而下降。	Loaded circuits heat one another. Cg depends on circuit or multicore-cable count, enclosure, layer arrangement, tray orientation and spacing. For the enclosed or bunched arrangement shown, the factors are 0.80 for two circuits and 0.65 for four, illustrating the reduction in available capacity.
Cable spacing 电缆间距	表中 spaced 表示相邻电缆表面净距至少为一个电缆外径 De。在表注所适用的水平排列中，水平净距大于 2De 可不作成组修正；该条件不能任意推广至其他围护或地沟情况。	In the table, spaced means a surface clearance of at least one cable diameter De. For the horizontal arrangements covered by its note, clearance exceeding 2De removes the grouping correction. Apply that concession only within the stated arrangement.

成组表注的影响 / Conditions attached to grouping factors

A5(3)的基本系数适用于同一截面的成组电缆。不同允许运行温度的电缆同组时，按组内最低允许温度确定额定值。仅在已知运行条件下，某电缆预计电流不超过其成组额定值 30%时，才可在计算其余电缆的成组系数时忽略该电缆；其自身仍须满足选线及保护要求。

The basic A5(3) factors apply to groups of one conductor size. Where cables have different permissible operating temperatures, base the group rating on the lowest. A cable expected, under known operating conditions, to carry no more than 30% of its grouped rating may be ignored when determining the factor for the remaining cables. It still needs appropriate sizing and protection.

例如普通 70°C PVC 在 45°C 环境下，A5(1)给出 0.79，而 A5(2)给出 0.91。应先确定保护器件的用途，再选温度表，随后按适用条件考虑 A5(5)。

For example, at 45°C, ordinary 70°C PVC has a factor of 0.79 in A5(1) and 0.91 in A5(2). Establish the protective-device duty before selecting the temperature table, then consider A5(5) where applicable.

4 隔热、保护器件及地沟修正 / Insulation, protection and trenches

因素 / Factor	定性解释	Qualitative explanation
Table A5(4) 隔热材料包围	建筑隔热材料阻碍散热。若电缆在隔热墙或天花内且一侧接触导热表面，应按 Method 4 及其条件选额定值。完全包围时，包围长度及材料导热性能决定进一步折减，不能仅因导体截面不变便沿用开放敷设额定值。	Thermal insulation restricts heat loss. For the stated insulated-wall or ceiling arrangement with one side in contact with a conducting surface, use Method 4 and its conditions. Total enclosure requires assessment of the enclosed length and insulation properties, rather than retaining an open-installation rating.
Table A5(5) 保护器件类型修正	过载保护额定值或整定值 In 必须与安装后的载流量协调。A5(5)对 BS 3036 半封闭熔断器给出 0.725，其他所列器件为 1.0。器件类型会影响所需表列载流量，不能仅比较负载电流与未修正的额定值。	Coordinate the overload-protection rating or setting In with installed cable capacity. A5(5) gives 0.725 for a BS 3036 semi-enclosed fuse and 1.0 for the other listed devices. The device type therefore affects the required tabulated capacity.
Table A5(6) 封闭地沟修正	封闭地沟内热量积聚。Method 18、19、20 须匹配沟体、盖板、排列、间距、数量和截面，并将 A5(6)用于相应 Method 12 或 13 的基础载流量。截面改变后须重新核对系数；不同温度电缆混组仍按最低允许运行温度。	Enclosed trenches retain heat. For Methods 18–20, match trench geometry, cover, cable layout, spacing, count and cross-section. Apply A5(6) to the corresponding Method 12 or 13 base capacity. Recheck the factor after a size change; mixed-temperature groups remain governed by the lowest permissible temperature.

隔热修正的适用范围 / Limits of the insulation factors

完全被隔热材料包围超过 0.5m 时，在没有更精确资料下，按开放直接固定 Method 1 载流量的 0.5 倍取值。短于 0.5m 时，A5(4)列出 50、100、200 及 400mm 对应 0.88、0.78、0.63 及 0.51；该表限用于不大于 10mm² 的导体，且隔热材料导热系数大于 0.04W/(m·K)。

For total enclosure exceeding 0.5m, use 0.5 times the Method 1 clipped-direct capacity in the absence of more precise information. For shorter lengths, A5(4) gives 0.88, 0.78, 0.63 and 0.51 at 50, 100, 200 and 400mm. Its stated scope is conductors up to 10mm² and insulation thermal conductivity greater than 0.04W/(m·K).

载流量及保护配合 / Capacity and protection coordination

$$I_b \le I_n \le I_z \qquad I_t \ge I_n \div (C_a \times C_g \times C_i \times C_d)$$

Ib 为设计电流，In 为过载保护额定值或整定值，Iz 为实际安装载流量，It 为表列载流量。第二式说明一般温度、成组、隔热和器件修正的选线关系。特殊地沟按 A5(6)及相关表注处理；各修正只在适用时采用，并避免重复计入同一热效应。

Ib is design current, In the overload-protection rating or setting, Iz installed capacity and It tabulated capacity. The second relation illustrates the usual temperature, grouping, insulation and device corrections. Treat special trenches using A5(6) and its notes. Apply only relevant corrections and avoid counting the same thermal effect twice.

5 安装方式及参考方法 / Installation and reference methods

Appendix 7 区分实际安装方式与查额定值所用的参考方法。工业厂房中常见的管槽、桥架、建筑孔洞及地沟，应按下列条件对应，再结合 Table A6(1)和适用修正系数。

Appendix 7 distinguishes the actual installation from the reference method used for rating. Match factory conduits, trunking, trays, building voids and trenches to the stated conditions before selecting the Table A6(1) column and applicable corrections.

因素 / Factor	定性解释	Qualitative explanation
Method 1 开放直接固定	带护套电缆直接固定于非金属表面，按 Method 1 查额定值。散热较开放，但仍须评估温度及成组。	Sheathed cables clipped directly to a non-metallic surface use Method 1. Cooling is relatively open, but temperature and grouping still require assessment.
Methods 3, 5, 7–10 一般管槽内	墙面或天花管道、非隔热砌体内管道，以及壁面、地板或踢脚线线槽，按所示条件采用参考 Method 3。围护限制散热。	The shown conduit, non-insulating masonry and wall, floor or skirting-trunking arrangements use reference Method 3. Their enclosure restricts cooling.
Methods 4, 6, 15 隔热墙或天花内	所示隔热墙或天花布置采用参考 Method 4，须符合一侧导热接触等图示条件。不能直接使用开放敷设栏位。	The specified insulated-wall or ceiling arrangements use reference Method 4, subject to their contact and heat-transfer conditions.
Method 11 有孔桥架	所示有孔桥架采用 Method 11。附图将开孔面积至少占 30%的桥架定义为有孔桥架；还须匹配单层、接触、方向等条件。	Use Method 11 for the stated perforated-tray arrangement. The figure defines a perforated tray as having holes occupying at least 30% of its surface; layout, touching and orientation also matter.
Methods 12–14 自由空气及梯架	Method 12 用于图示带护套单芯电缆，Method 13 用于图示多芯布置；悬挂方式 14 按 12 或 13 处理。须满足间距及支承遮挡条件，包括图示支承金属占平面面积小于 10%。	Method 12 covers the shown sheathed single-core arrangement and Method 13 the multicore arrangement. Method 14 uses 12 or 13 as appropriate. Match the stated clearances and supporting-metalwork restriction, including less than 10% plan-area obstruction where specified.
Method 16 建筑孔洞内	非隔热结构孔洞按尺寸选参考方法：图示直径不大于 5De 或周长不大于 20De 时采用 4，超过相应界限时采用 3；周长超过 60De 则按适合的 18–20 处理。De 的组缆定义也须依表注。	For the stated non-insulating building void, the small-void limits are 5De in diameter or 20De in perimeter for Method 4; beyond the relevant limit, Method 3 applies. Perimeter exceeding 60De invokes an appropriate Method 18–20. Use the stated group definition of De.
Method 17 开放或通风地沟	按图示支承及间距采用 Method 12 或 13。通风及净距条件必须实际满足。	An open or ventilated trench uses Method 12 or 13 with the specified support and spacing. The ventilation and clearances must actually be provided.
Methods 18–20 封闭地沟	采用所示地沟布置及 A5(6)，核对盖板、尺寸、间距和组数。该类布置须作专用修正，不能仅因电缆装在梯架上便按无围护的额定值使用。	Use the depicted enclosed-trench arrangement and A5(6), checking cover, dimensions, spacing and group count. The presence of ladder supports does not remove the enclosed-trench correction.

6 Table A6(1)：电缆规格、载流量与压降
Cable specification, current rating and voltage drop

本表主要适用于铜导体、单芯、PVC 绝缘、非铠装电缆，可带或不带护套。载流量以环境 30°C、导体 70°C 为参考，必须按电缆截面、相制、导体数量及参考 Method 选正确栏位。

The main table covers single-core, PVC-insulated, non-armoured copper cables, with or without sheath. Its current ratings use 30°C ambient and 70°C conductor temperature. Select the correct column for size, phase arrangement, conductor count and reference Method.

因素 / Factor	定性解释	Qualitative explanation
Cable type and table notes 类型及表注	铝、多芯、XLPE 或铠装结构须采用适用额定资料。表注仅允许指定 BS 6004 柔性电缆及 BS 6231 的 85／90°C 耐热 PVC 在固定安装时使用第 2–5 栏载流量，不能将该许可扩展至所有栏位。	Use appropriate rating data for aluminium, multicore, XLPE or armoured constructions. The note permits only the specified BS 6004 flexible and BS 6231 85/90°C heat-resisting PVC cables to use columns 2–5 for fixed installations; that permission is column- and application-specific.
Current rating and layout 载流量及排列	相同截面并无单一固定载流量。Method、单相或三相、接触或分隔及平面或三叶排列都会影响额定值。表中“—”没有提供该组合的数据，须另取适用资料。	A conductor size has no single universal current rating. Method, phase arrangement, spacing and flat or trefoil formation affect the rating. A dash supplies no rating for that combination; obtain applicable data rather than borrowing another column.
Length and voltage drop 长度及压降	较长线路和较大电流通常增加压降。压降页以 mV/A/m 列值，须匹配直流、单相或三相及排列。较大截面列出 r、x、z，分别表示电阻、电抗及阻抗；交流负载的功率因数也须纳入判断。	Longer routes and higher current generally increase voltage drop. Use the mV/A/m column for the correct d.c. or a.c. phase and formation. Larger sizes distinguish resistance r, reactance x and impedance z; load power factor also requires consideration.

相同 4mm² 截面的定性比较 / Comparing the same 4mm² size

参考方法 / Reference method	三相基准载流量 Three-phase base rating	含义 / Meaning
Method 4	24A	隔热墙内管道，散热受限 / Conduit in a thermally insulating wall.
Method 3	28A	一般墙面管道或线槽 / Conduit on a wall or trunking.
Method 1	33A	直接固定，散热较开放 / Clipped direct, with more open cooling.

这些是三相基准值，使用前仍须作适用修正。同一 4mm² 在压降页的三相值为 9.5mV/A/m，单相为 11mV/A/m。间距增大虽可改善散热，但表注指出超过 Method 12 规定的间距会使压降增大，须同时检查两者。

These are three-phase base ratings before correction. The 4mm² voltage-drop entries are 9.5mV/A/m for three-phase and 11mV/A/m for single-phase. Greater spacing may improve cooling, but spacing beyond Method 12 increases voltage drop according to the note. Check both effects.

上游与下游压降应合并检查；电动机回路还应核查启动期间的电压下降。

Check upstream and downstream voltage drop together. Motor circuits also require a starting-voltage check.

7 工业环境中的其他约束 / Additional industrial constraints

因素 / Factor	定性解释	Qualitative explanation
Fault withstand and protection 故障承受与保护	导体及绝缘须承受故障切除前的热应力，保护器件分断能力须足够。还须检查故障环路阻抗、保护导体连续性及切除条件；载流量合格并不代表故障保护已合格。	The conductor and insulation must withstand fault energy until disconnection, and the protective device needs adequate breaking capacity. Check loop impedance, protective-conductor continuity and disconnection conditions in addition to normal current capacity.
External influences 外部环境	潮湿、水、油、腐蚀性气体、化学品、粉尘和外部热源影响绝缘及护套寿命。须选择相容材料及密封防护，增大导体截面不能替代环境适用性。	Moisture, water, oil, corrosive substances, chemicals, dust and external heat affect insulation and sheath life. Select compatible materials and sealing; a larger conductor does not replace environmental suitability.
Mechanical stresses 机械应力	冲击、磨损、振动、拉力及移动影响铠装、管槽和支承选择。移动机械应采用适合反复弯曲的电缆；弯曲半径、牵引和固定条件也须满足。	Impact, abrasion, vibration, tension and movement govern armour, containment and support. Moving machinery needs cables suitable for repeated flexing. Bending, pulling and fixing requirements must also be met.
Fire and explosion conditions 火灾及爆炸条件	应考虑火焰蔓延及烟气；消防或应急线路还须考虑火灾时的供电连续性。可燃气体或粉尘区域的电缆、进线装置和布线系统须适合相应危险条件。	Consider flame spread and smoke, and continuity of supply for fire-service or emergency circuits. In flammable-gas or dust areas, the cable, entries and wiring system must suit the hazardous conditions.
Terminations and maintenance 端接及维修	导体材料、截面及外径须与端子、接线耳、格兰头和管槽容量配合。接头接触不良会局部过热；布线应便于检查、测试及更换，并满足端子温度限制。	Conductor material, size and cable diameter must suit terminals, lugs, glands and enclosure capacity. Poor contacts cause local heating. Provide access for inspection, testing and replacement, and respect terminal temperature limits.
Power loss and future loading 损耗及未来负荷	长时间带载的工业线路应考虑 I^2R 损耗和运行费用。合理增大截面可降低电阻及损耗；扩建或新增成组回路后，应重新检查负载、修正系数、压降及保护配合。	For long operating hours, assess I^2R losses and running cost. A larger conductor can reduce resistance and losses. Following expansion or added grouped circuits, reassess loading, corrections, voltage drop and protection coordination.

综合选择 / Integrated selection

最终规格应同时适合工业环境、保护器件和全部安装条件，并通过压降、故障及损耗检查。若某一条件不满足，应增大截面、改善间距、通风或路线，或选用更适合的电缆结构，再重新核查相关表格和修正系数。

The final cable must suit the industrial environment, protective arrangement and all installation conditions, while meeting voltage-drop, fault and loss checks. If a condition is not satisfied, increase size, improve spacing, ventilation or routing, or select a more suitable construction, then recheck the relevant rating and correction data.